

Lecture 4: Point Processing - Part 2

Histogram and Normalized Histogram

1. Histogram

The histogram is defined as:

$$h_r(r_k) = n_k = |\{I(x, y) | I(x, y) = r_k\}| \quad r_k = 0, 1, 2, \dots, 255$$

Meaning:

- $I(x, y)$ = intensity value at pixel location (x, y)
- r_k = a specific gray level
- n_k = number of pixels whose intensity equals (r_k)
- $|\cdot|$ = number of elements in the set

So, The histogram counts how many pixels have intensity (r_k) .

2. Normalized Histogram

$$f_r(r_k) = \frac{n_k}{N_T}$$

Where:

- n_k = number of pixels with intensity (r_k)
- N_T = total number of pixels in the image

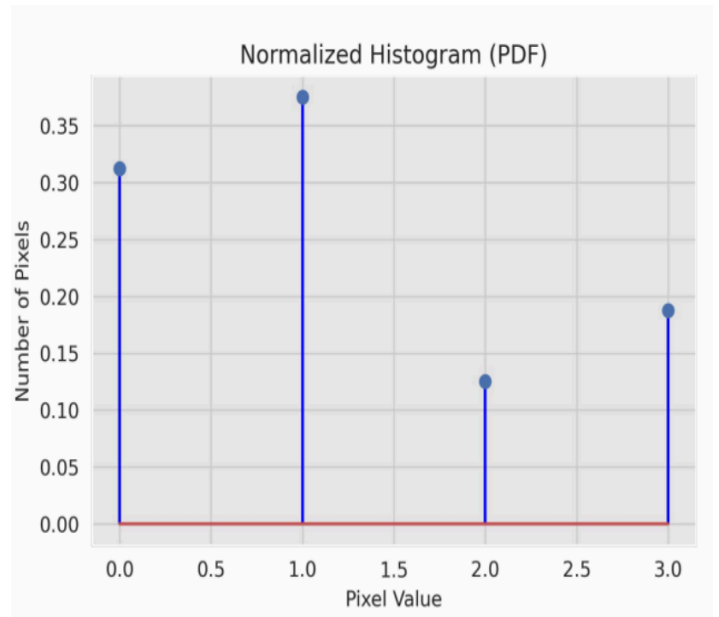
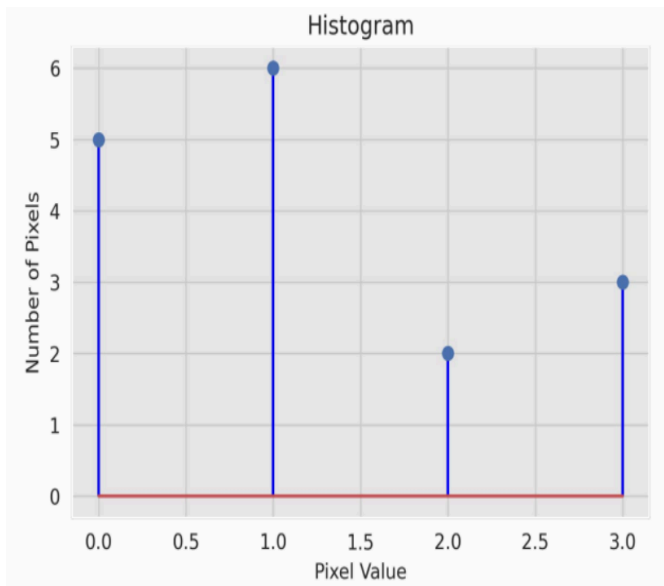
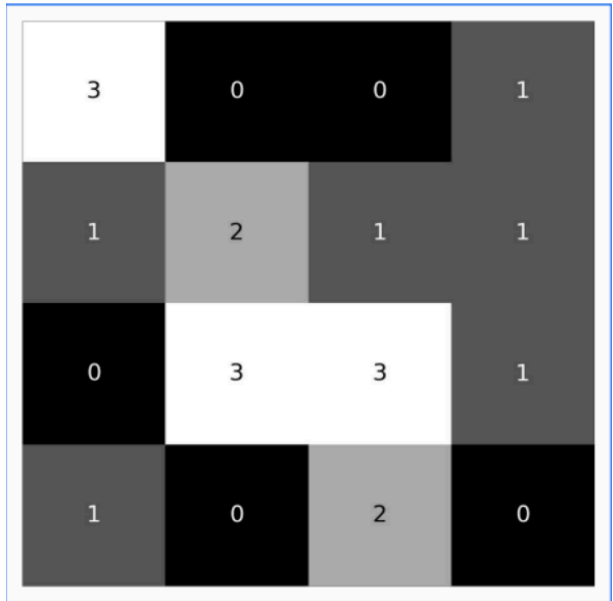
Meaning:

It gives the **percentage (or fraction)** of pixels having intensity (r_k) .

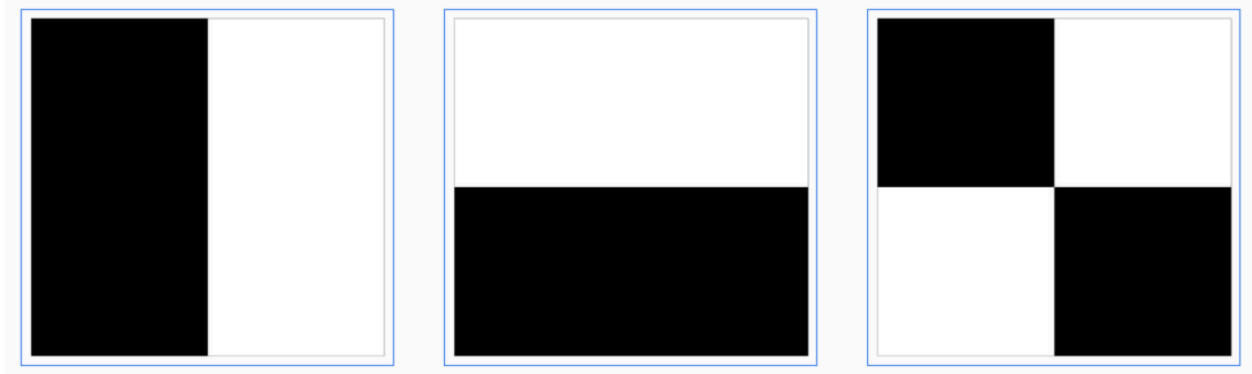
Histogram

Example: A (4 x 4) image with 2 bit or 4 quantization levels

r_k	n_k	fr(r_k)
0	5	$5/16 = 0.31$
1	6	$6/16 = 0.38$
2	2	$2/16 = 0.12$
3	3	$3/16 = 0.19$



All of these images have the exact same histogram (equal number of black and white).



The second image is a pixel permutation of the first image, hence the same histogram.



In the first image, most of the maximum intensity values are close to black, meaning many pixel values fall within the range 0–50 (as shown in the third image). Very few pixels have intensity values between 200–250, and pixels with value 255 are the least frequent. Therefore, the histogram is heavily concentrated in the lower gray-level range.

If we rearrange the positions of the pixels in the first image, it may visually resemble the second image. This process is called **pixel permutation**. However, since the gray-level values themselves are not changed- only their positions- the histogram remains exactly the same for both images.

This demonstrates that histograms are not unique; different images can share the same histogram as long as the pixel intensity values remain unchanged, even if their spatial arrangement differs.

Cumulative Histogram & CDF

1. Cumulative Histogram

For a given gray level (r):

Mathematically:

$$H_r(r) = n_k = |\{I(x, y) | I(x, y) \leq r\}| = \int_{-\infty}^r h_r(x) dx \quad r \in \mathbb{R}$$

It means: Number of pixels with intensity $\leq r$

Where:

- $I(x, y)$ = pixel intensity
- n_k = number of pixels with intensity $\leq (r)$

Relation with Histogram

Cumulative histogram = sum of histogram values up to r

2. Normalized Cumulative Histogram (CDF)

If we divide the cumulative histogram by total pixels (N_T), we get:

$$F_r(r) = \frac{n_k}{N_T} = \int_{-\infty}^r f_r(x) dx$$

It gives: Probability that a randomly chosen pixel has value $\leq r$

Why is it called CDF?

Because it satisfies properties of a **Cumulative Distribution Function**:

- $0 \leq F_r(r) \leq 1$

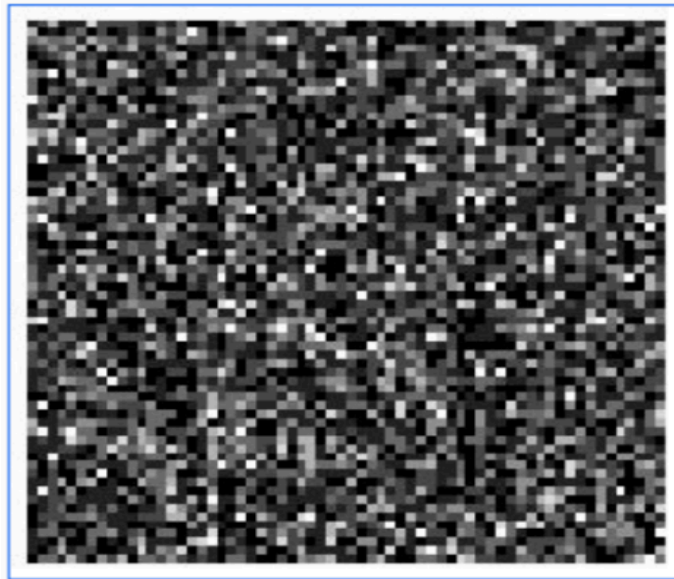
- It is **monotonically increasing**
- $Fr(\infty)=1$

Important:

- Histogram $\rightarrow$ PDF
- Cumulative Histogram $\rightarrow$ CDF
- CDF is used directly in **histogram equalization transformation function**

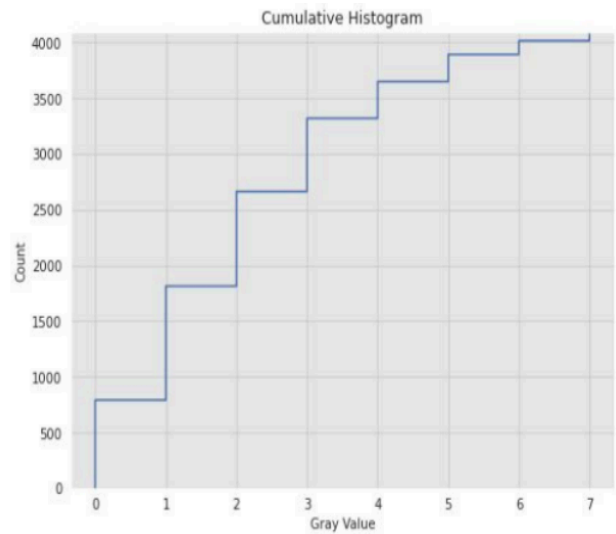
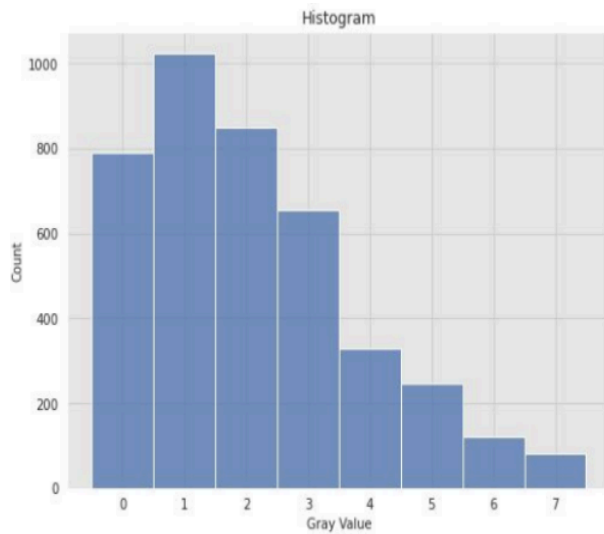
Cumulative Histogram

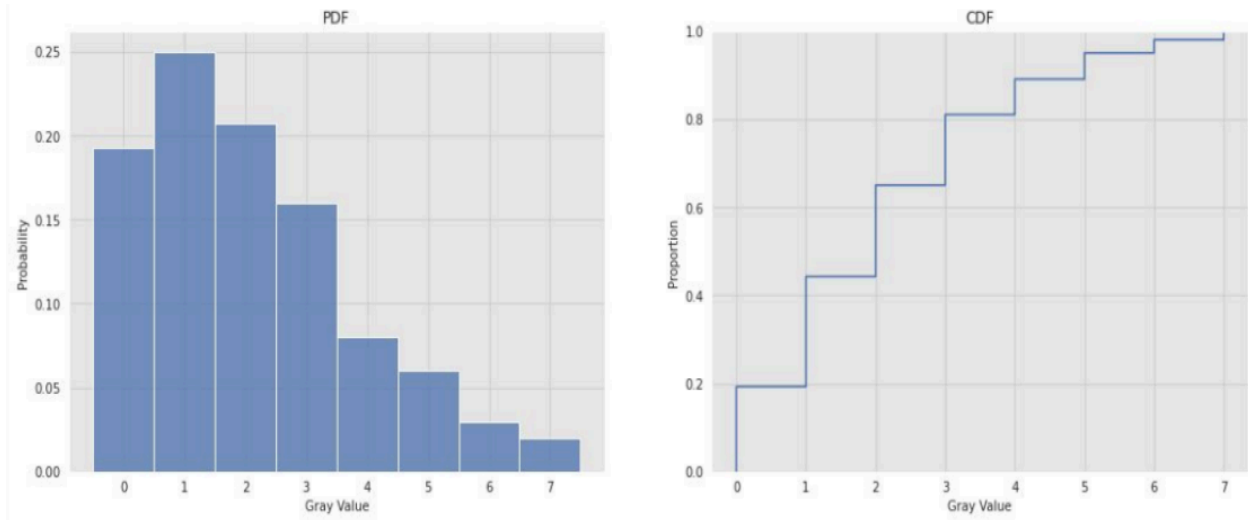
Example: A random (64 x 64) image with 3 bit or 8 quantization levels.



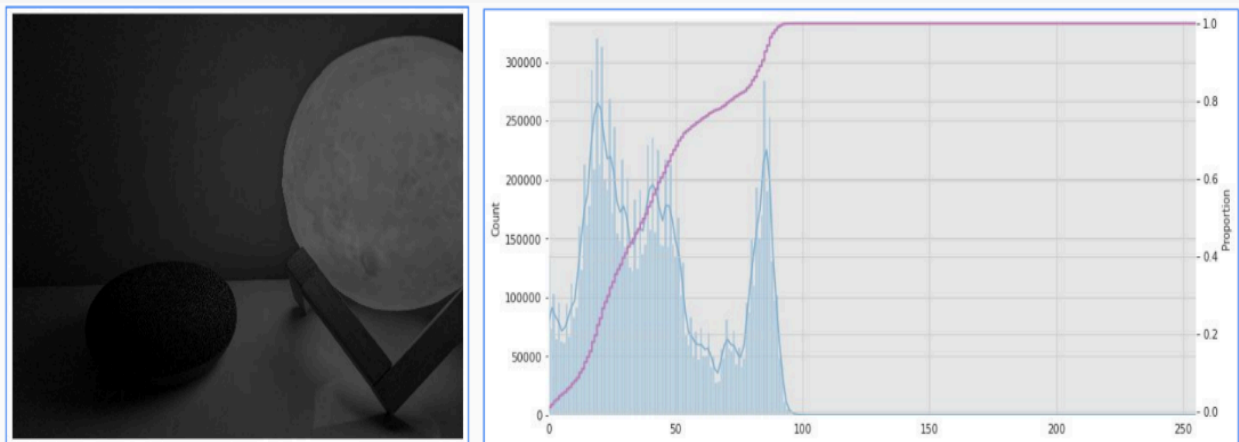
r_k	$h(r_k)$	$H(r_k)$	$f(r_k)$ [PDF]	$F(r_k)$ [CDF]
0	790	790	$790/(64 \times 64) = 0.19$	0.19

1	1023	$790+1023=1813$	$1023/4096=0.25$	$0.19+0.25=0.44$
2	850	$1813+850=2663$	$850/4096=0.21$	$0.44+0.21=0.65$
3	656	$2663+656=3319$	$656/4096=0.16$	$0.65+0.16=0.81$
4	329	$3319+329=3648$	$329/4096=0.08$	$0.81+0.08=0.89$
5	245	$3648+245=3893$	$245/4096=0.06$	$0.89+0.06=0.95$
6	122	$3893+122=4015$	$122/4096=0.03$	$0.95+0.03=0.98$
7	81	$4015+81=4096$	$81/4096=0.02$	$0.98+0.02=1.00$





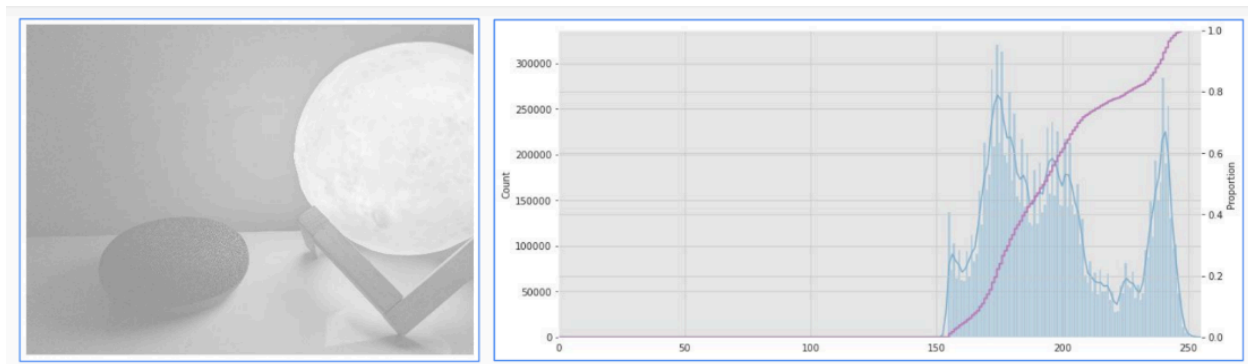
Histogram of a dark image



The blue bars represent the **histogram**, and the violet line represents the **CDF (Cumulative Distribution Function)**.

If the CDF reaches 1 (or 100%) at gray level 100, it means that all pixel intensities are less than or equal to 100. In other words, there are no pixels with values greater than 100. This indicates that the histogram is concentrated in the lower gray-level range. Since most of the pixel values are small (closer to 0, which represents black), the image appears dark.

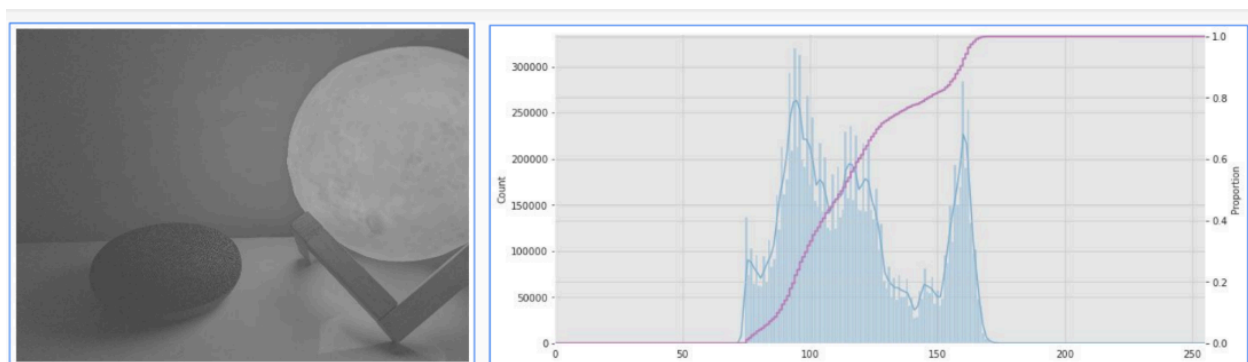
Histogram of a bright image



In a bright image, most pixel values lie in the range **155 to 255**.

If the CDF value at gray levels 100 or 150 is **0**, it means there are **no pixels** with intensity less than or equal to 100 or 150. In other words, all pixel values are greater than 150. This shows that the histogram is concentrated in the higher gray-level range. Since most pixel intensities are close to 255 (white), the image appears bright.

Histogram of a low contrast image



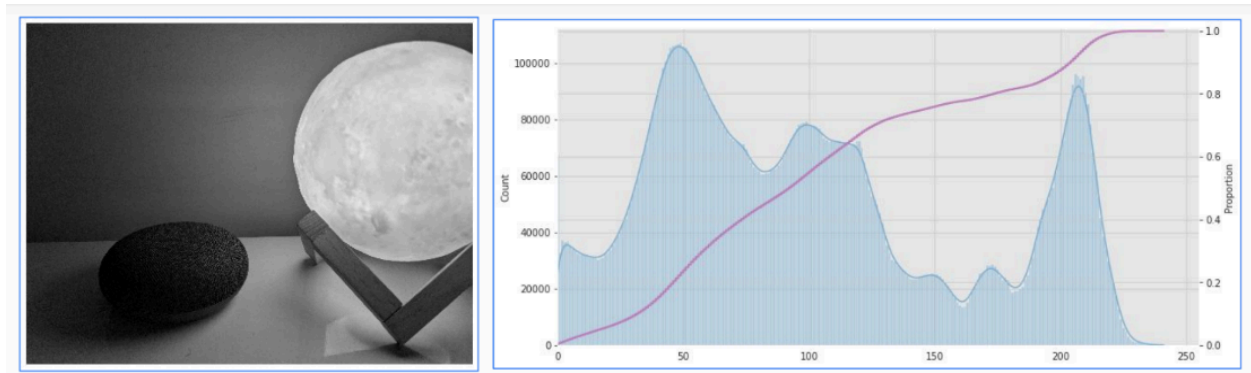
If most of the pixel values are concentrated between **75 and 175**, it means the intensities are limited to the middle gray-level range.

Since the pixels are neither very dark (close to 0) nor very bright (close to 255), the image does not have strong black or white regions. As a result, the contrast is low.

This type of image often looks **“washed out”** because it lacks deep shadows and bright highlights. The histogram being concentrated in the mid-range gray values

clearly indicates a **low-contrast image**, such as some satellite images where intensity variations are small.

Histogram of a high contrast image

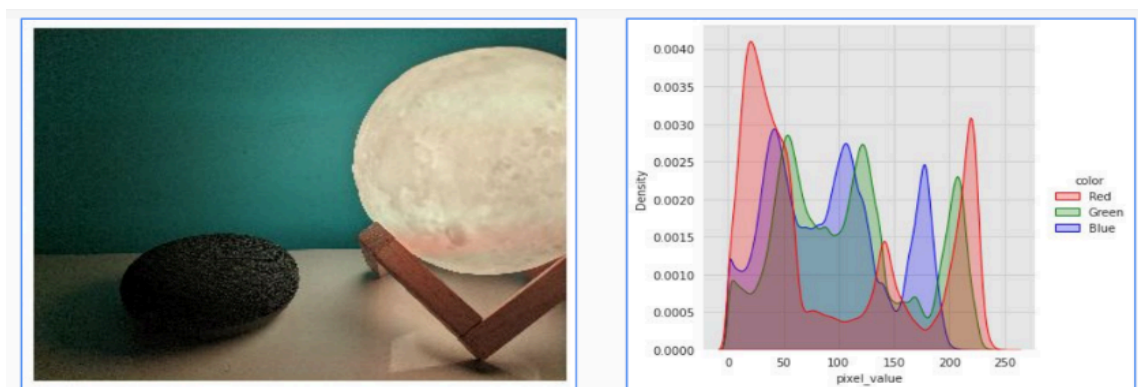


If the pixel values are distributed across the entire range **0–255**, it means the gray levels are spread out rather than concentrated in a small region.

If this distribution is approximately **uniform**, then all intensity levels occur with nearly equal frequency. This indicates a **high-contrast image**, because the image contains very dark, mid-tone, and very bright pixels.

The contrast can sometimes be enhanced further by **histogram equalization** (often described as “flattening” the histogram). This process redistributes the pixel intensities to make the histogram more uniform, which can improve visual contrast- especially if the original distribution was not perfectly uniform.

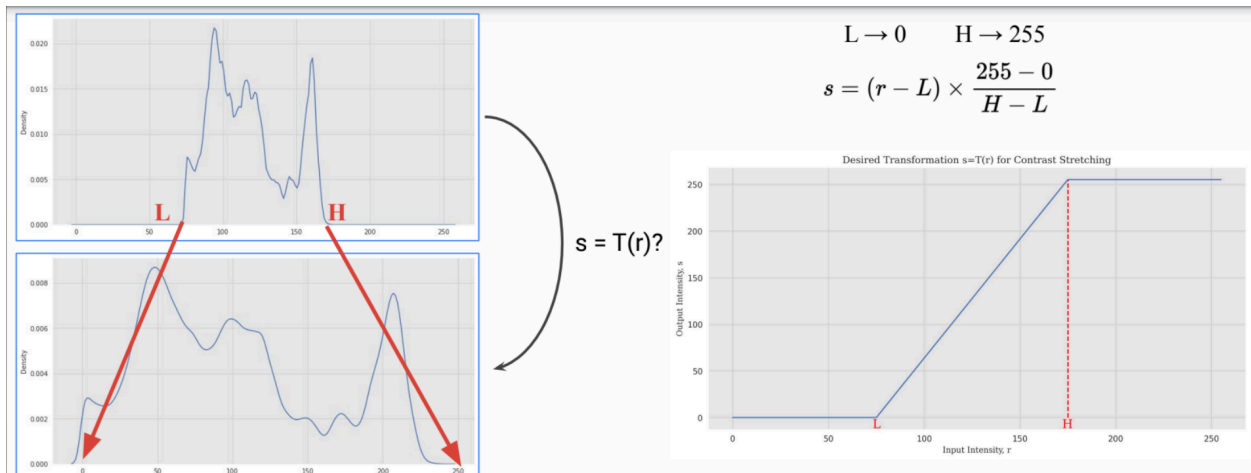
Histograms of a color image



The histogram of a color image consists of three separate histograms, one for each color channel: red, green, and blue. Each channel's pixel distribution is analyzed individually. In this particular image, the red channel values are concentrated in the lower range, indicating that the image contains relatively few red pixels.

Contrast Stretching

Histogram Point of View



Contrast Stretching (Histogram Point of View)

- In the original image, most pixel intensities lie between **L** and **H** (narrow range).
- This means the image has **low contrast**.

Idea

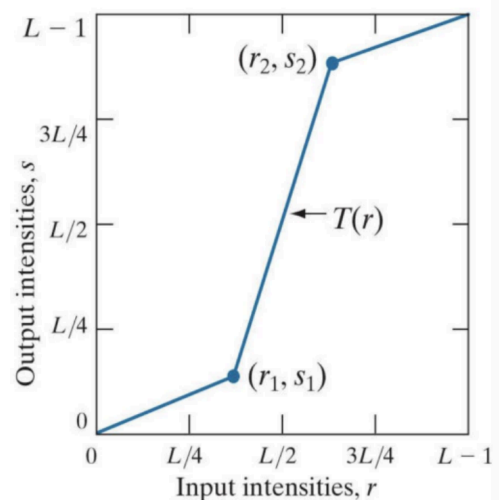
Stretch the intensity range $[L, H] \rightarrow [0, 255]$ to increase contrast.

Transformation

$$s = (r - L) \times \frac{255 - 0}{H - L}$$

Where

- **r** = input intensity
- **s** = output intensity

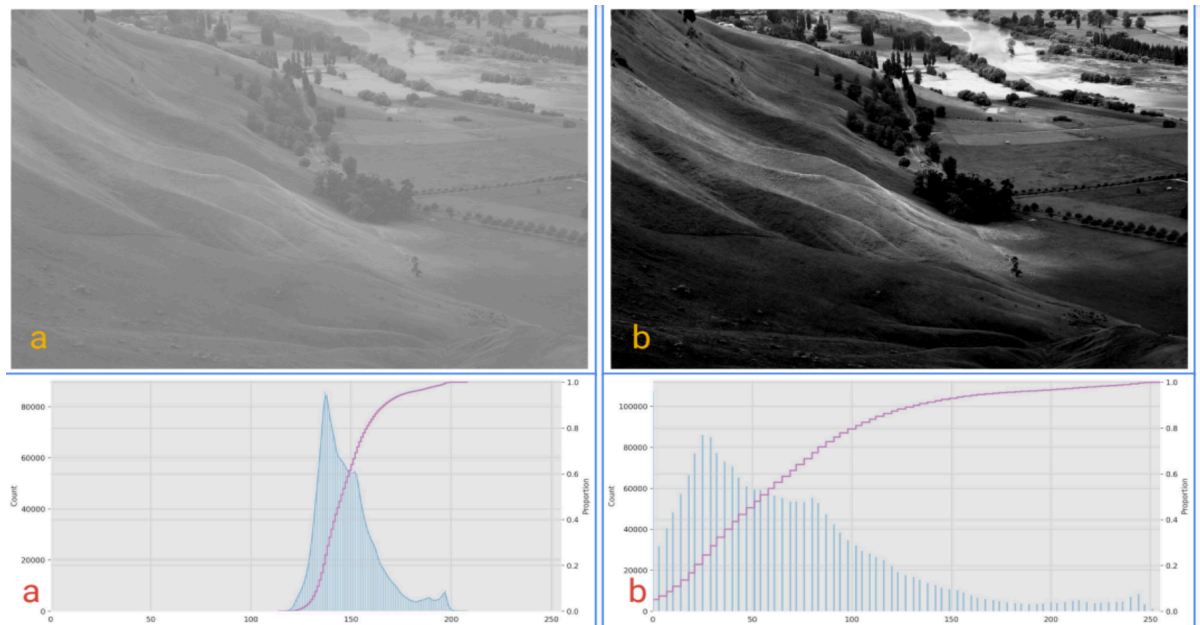


- **L** = lowest useful gray level
- **H** = highest useful gray level

Result

- $r \leq L \rightarrow s = 0$ (black)
- $L < r < H \rightarrow$ **stretched linearly**
- $r \geq H \rightarrow s = 255$ (white)

So the histogram spreads across **0–255**, making the image **higher contrast**.



Explanation of the Figure (Contrast Stretching)

Left side (a):

- The original image looks **washed out / low contrast**.
- The histogram shows most pixel values **clustered in a small range** (around mid-gray).
- Because intensities are not spread from **0–255**, the image lacks contrast.

Right side (b):

- After **contrast stretching**, the image becomes **clearer and sharper**.

- The histogram spreads over a **wider range (0–255)**.
- Dark pixels become darker and bright pixels become brighter.

Idea:

Contrast stretching **expands the narrow intensity range of the original image to the full range (0–255)**, improving visibility and contrast.

Histogram Equalization

We can do better by Histogram Equalization.

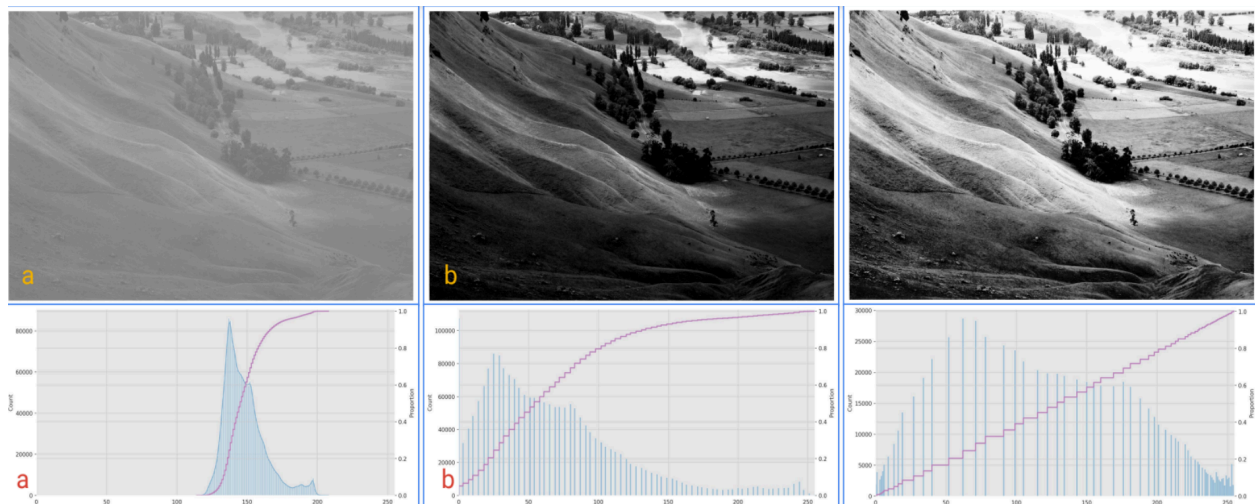
Contrast stretching only **spreads the gray levels linearly** between (L) and (H).
But the histogram may still **not be evenly distributed**.

Histogram Equalization improves it further by:

- **Redistributing pixel intensities** using the **CDF**.
- Making the histogram **more uniform over 0–255**.
- This **enhances contrast more effectively**, especially in images where pixels are heavily concentrated in a small range.

So, histogram equalization usually gives better contrast enhancement than simple contrast stretching.

Histogram Equalization (Figure Explanation)



a - Original Image

- The image looks **low contrast**.
- Histogram is **narrow and concentrated** around some gray levels.
- CDF (purple curve) is **steep in a small region**, meaning many pixels have similar intensities.

b - After Contrast Stretching

- Image contrast **improves**.
- Histogram spreads more, but **still not uniform**.
- CDF becomes **smoother but not linear**.

c - After Histogram Equalization

- Contrast is **further improved**.
- Histogram becomes **more evenly distributed** across 0–255.
- CDF becomes **almost a straight line**.

Key idea:

If the CDF is linear (straight line), it means gray levels are uniformly distributed, so the image uses the full intensity range, giving good contrast.

Histogram Equalization – Motivation (Entropy)

$$\text{Entropy}(I) = \sum_{r_k} -f(r_k) \log f(r_k)$$

Where

- $f(r_k)$ = probability of gray level (r_k) (normalized histogram).

Meaning

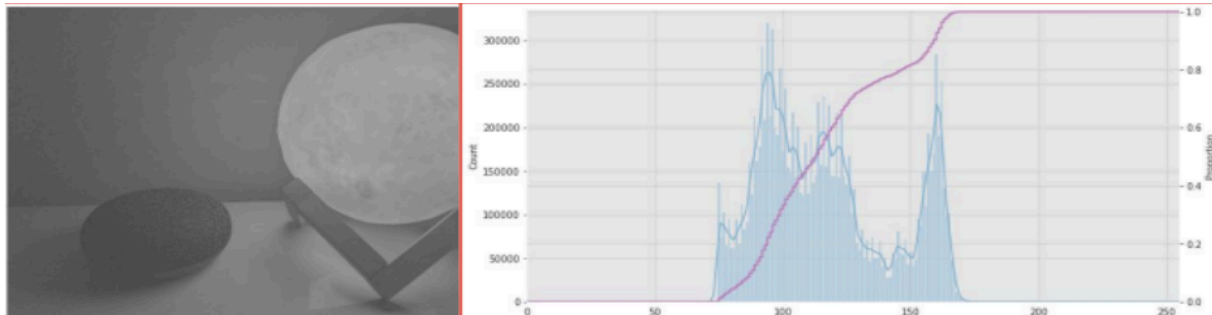
- Entropy measures uncertainty or randomness of pixel intensities.
- Higher entropy → more information in the image.

Key ideas

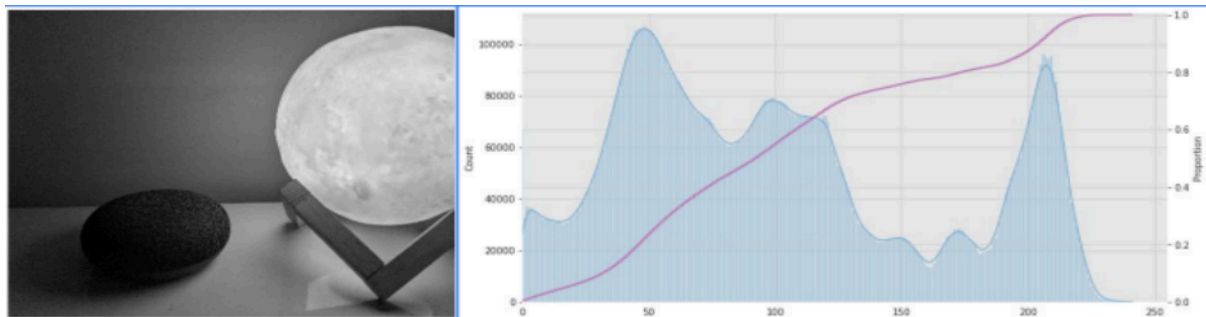
- **Certain events (pixels concentrated at few levels)** → low entropy → low contrast.
- **Uncertain events (pixels spread over many levels)** → high entropy → more information.

Relation to Histogram Equalization

- **Uniform histogram** → pixels spread across all gray levels.
- This gives **high entropy and high contrast**, which is desirable.

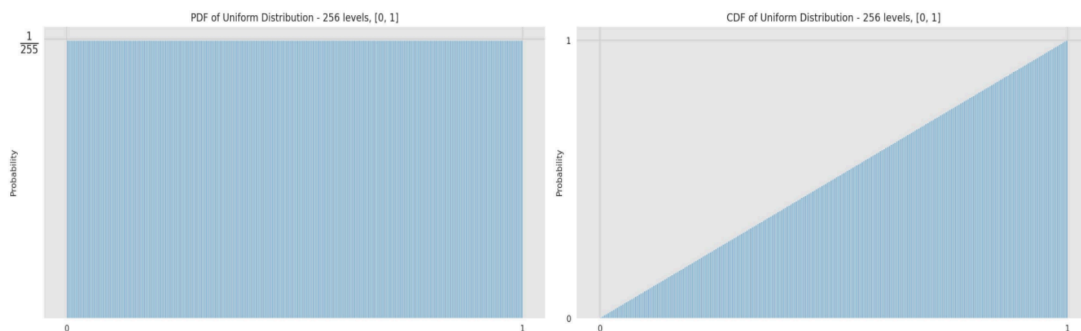


Entropy = 6.34



Entropy = 7.64

Motivation - Uniform Distribution



Idea

If an image is **washed out (low contrast)**, we try to make the **histogram uniform** using **Histogram Equalization**.

PDF

For a perfectly equalized image, the **PDF becomes flat (uniform distribution)**:

$$f(r_k) = \frac{1}{L - 1}$$

where

- **L** = number of gray levels (usually 256)

This means **all gray levels occur with equal probability**.

CDF

The **CDF becomes linear** between the minimum and maximum intensity.

- Straight line
- **Slope = 1**

Meaning

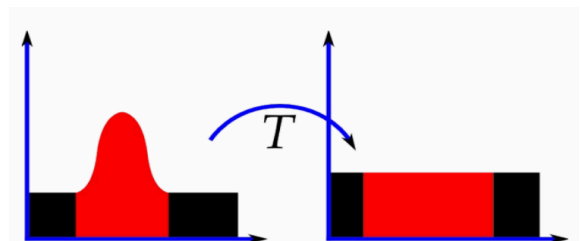
- Flat PDF → pixels evenly distributed
- Linear CDF → full intensity range used
- Result → higher contrast image.

Basic Idea of Histogram Equalization

Goal

Find a transformation:

$$s = T(r)$$



so that the **output histogram becomes uniform (flat PDF)** and the **CDF becomes linear**.

Observation (normalized gray levels 0–1):

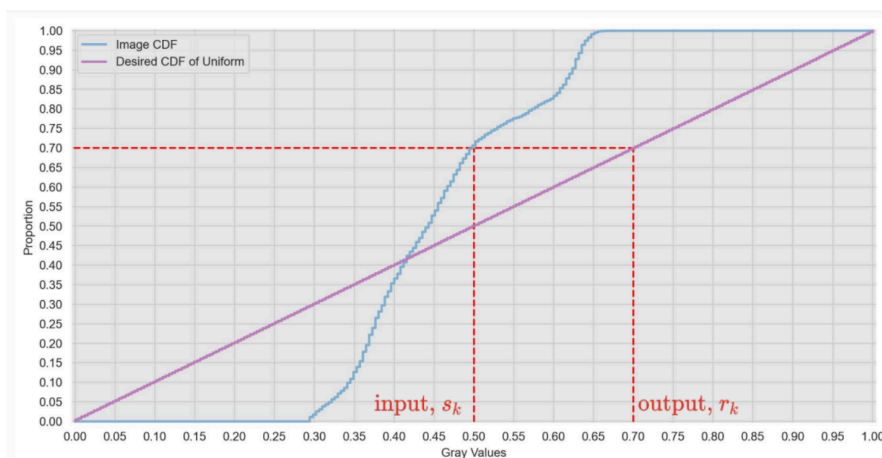
- In an **ideal uniform distribution**, **y% of pixels have values $\leq y$** .
(Because the **CDF is linear**.)
- Reverse idea:
If **y% of pixels are smaller than a pixel**, then that pixel **should have intensity y**.

Idea (Assuming normalized gray values $[0, 1]$)

- For a given gray value x , we know the percentage of pixels with values $\leq x$ (CDF)
- Hence, map x to its CDF value to get a flat histogram/linear CDF.

Result:

Mapping each pixel to its **CDF value** spreads intensities across the range, producing a **flat histogram and improved contrast**.



Hence,

$$s_k = T(r_k) = \text{CDF}_r(r_k) = F_r(r_k)$$

For L quantization levels,

$$s_k = T(r_k) = \text{round}((L - 1)F_r(r_k))$$

The image explains the fundamental concept of **Histogram Equalization**, an image processing technique used to improve contrast.

Core Concept

The goal is to transform the intensity values of an image so that the output has a **uniform distribution** (meaning all gray levels are used equally).

- **Blue Line (Image CDF):** Represents the Cumulative Distribution Function of the original image. Notice it is steep in the middle, meaning most pixels are concentrated in a narrow range of gray values (low contrast).
- **Pink Line (Desired CDF):** Represents a perfectly uniform distribution, which appears as a straight diagonal line.
- **The Mapping:** To "equalize" the image, an input gray value (s_k) is mapped to a new output value (r_k) by finding where its CDF value hits the ideal uniform line.

The transformation function $T(r_k)$ maps the input intensity to the output:

- **Continuous Form:** The output value is simply the CDF of the input:

$$s_k = T(r_k) = \text{CDF}_r(r_k)$$
- **Discrete Form (for Digital Images):** To fit into specific quantization levels (like 0–255), the result is scaled and rounded:

$$s_k = \text{round}((L - 1) F_r(r_k))$$

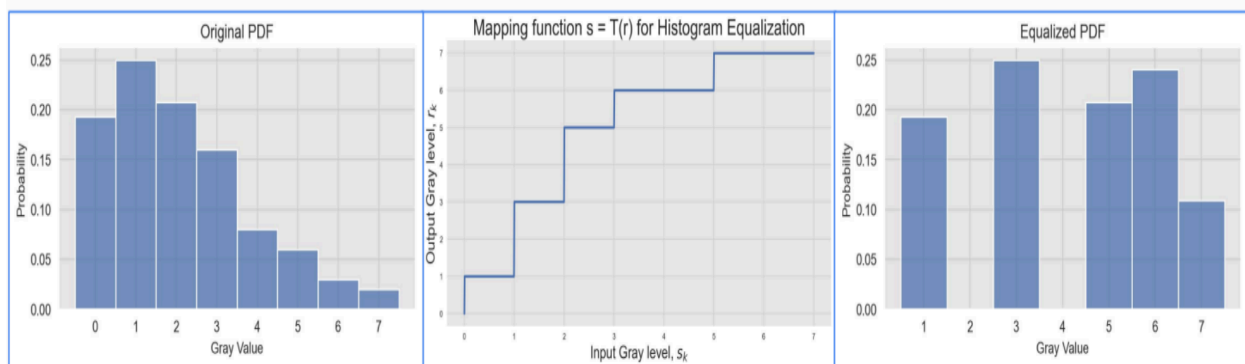
Where L is the number of possible intensity levels.

Numerical Example

A random (64 x 64) image with 3 bit or 8 quantization levels ($L = 8$)

rk (Input)	h(rk) (Count)	f(rk) [PDF]	H(rk) (Sum)	F(rk) [CDF]	sk=round(7×F(rk))
0	790	0.19	790	0.19	round(7x0.19)=1

1	1023	0.25	1813	0.44	$\text{round}(7 \times 0.44) = 3$
2	850	0.21	2663	0.65	$\text{round}(7 \times 0.65) = 5$
3	656	0.16	3319	0.81	$\text{round}(7 \times 0.81) = 6$
4	329	0.08	3648	0.89	$\text{round}(7 \times 0.89) = 6$
5	245	0.06	3893	0.95	$\text{round}(7 \times 0.95) = 7$
6	122	0.03	4015	0.98	$\text{round}(7 \times 0.98) = 7$
7	81	0.02	4096	1.00	$\text{round}(7 \times 1.00) = 7$



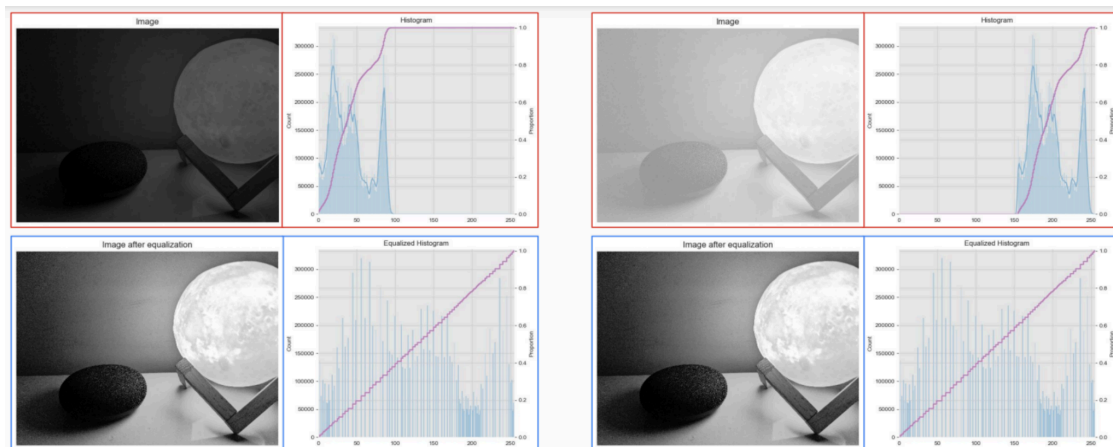
Visualization of the Transformation

The process is visualized through the following stages:

- **Original PDF:** Shows a "dark" image where most pixels are concentrated in the lower gray values (0, 1, and 2).

- **Mapping Function:** A step-like function $s = T(r)$ that maps the input intensities to the output intensities based on the cumulative probabilities.
- **Equalized PDF:** The resulting distribution where the gray levels are "stretched" across the available range, increasing visual contrast.

Examples



This image illustrates **Histogram Equalization**, a fundamental digital image processing technique used to improve contrast. By spreading out the most frequent intensity values, it allows areas of lower local contrast to gain a higher contrast.

Here is a breakdown of what is happening in the examples:

How It Works

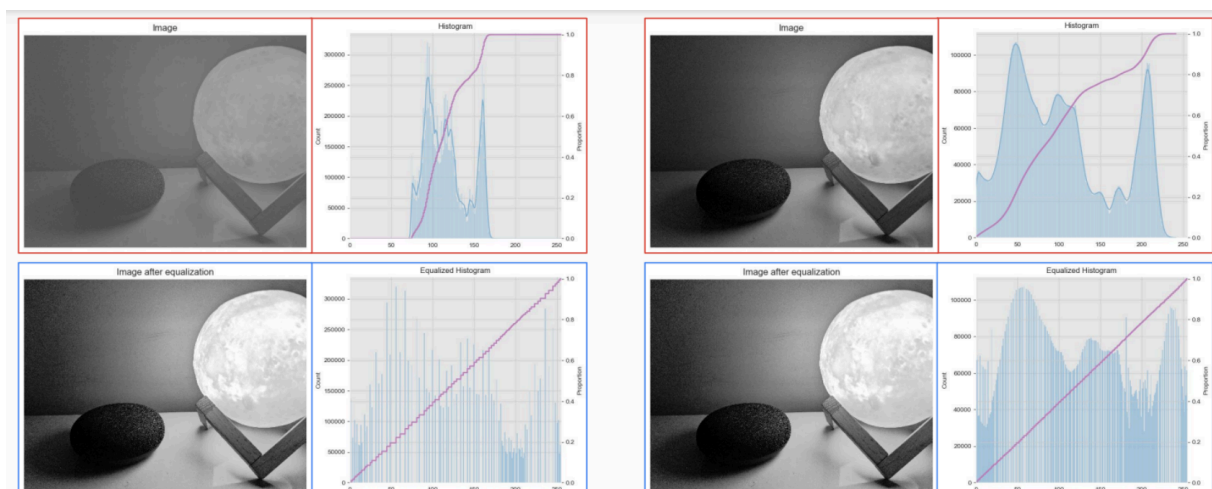
The goal of histogram equalization is to take an image with a "bunched up" histogram and transform it so that the intensity levels are distributed more uniformly across the entire range (usually 0 to 255).

- **The Problem:** In the top row of both examples, the original images are either too dark (left) or too "washed out" (right). Their histograms show that all the pixels are crowded into a small range of gray levels.
- **The Transformation:** The algorithm uses the **Cumulative Distribution Function (CDF)**- represented by the purple line in the graphs- to map the original pixel values to new ones.
- **The Result:** Notice the bottom histograms. The "Equalized Histograms" are much wider, and the purple CDF line becomes a straight diagonal. This

indicates that the pixels are now evenly distributed, resulting in the much clearer, higher-contrast images on the bottom.

Analysis of the Two Cases

Feature	Case 1 (Left)	Case 2 (Right)
Initial State	Underexposed: The histogram is skewed to the left (darker values).	Overexposed: The histogram is skewed to the right (lighter values).
Visual Effect	The background and the object on the left are nearly invisible.	The image looks "foggy" or low-detail because everything is bright.
After Equalization	Details in the shadows (like the texture of the round object) become visible.	The "fog" is removed, and shadows are deepened, making objects stand out.



This example further demonstrates the power of **Histogram Equalization** in enhancing image contrast, particularly for images that are "flat" or have intensities concentrated in a narrow range.

Contrast Comparison

The images in the top row (red boxes) exhibit poor contrast, but for different reasons than the previous examples:

- **Left Example (Low Contrast/Gray):** The original image is very "gray." Its histogram shows that almost all pixels are clustered in the middle intensities (roughly between 75 and 175). There are no true blacks or true whites, making the image look dull.
- **Right Example (High Initial Variance):** This image starts with a broader distribution than the left one, but it still lacks a full range. Equalization stretches these peaks to occupy the entire 0-255 spectrum.

The Transformation Process

Histogram equalization uses the **Cumulative Distribution Function (CDF)**- the purple line- to re-map the image.

1. **Original CDF:** In the top graphs, the purple line is steep where the pixel "count" is high and flat where it is low. This steepness indicates a high concentration of pixels in a small range.
2. **Linearization:** The goal of the equalization algorithm is to make this purple line a straight diagonal from (0,0) to (255,1.0).
3. **Equalized Result:** In the bottom blue boxes, the "Equalized Histogram" shows that the pixel intensities have been "pushed" apart. While the general shape of the peaks remains similar, they are now spread across the full horizontal axis.

Wrong method - channel wise equalization



Correct method - RGB to LAB, equalize L, LAB to RGB



The image explains how to perform **Histogram Equalization** on color images without distorting the actual colors.

The Problem: Channel-wise Equalization (Wrong Method)

When we equalize the Red, Green, and Blue (RGB) channels separately:

- **Intensity vs. Color:** In RGB, the brightness information is tied to the color information.
- **Color Shifting:** Changing the distribution of one channel independently of the others changes the ratio between R, G, and B. This results in "fake" colors and a noisy, unrealistic look (as seen in the bottom-left image).

The Solution: Using the LAB Color Space (Correct Method)

To fix the brightness while keeping colors natural, we must separate **what** the color is from **how bright** it is.

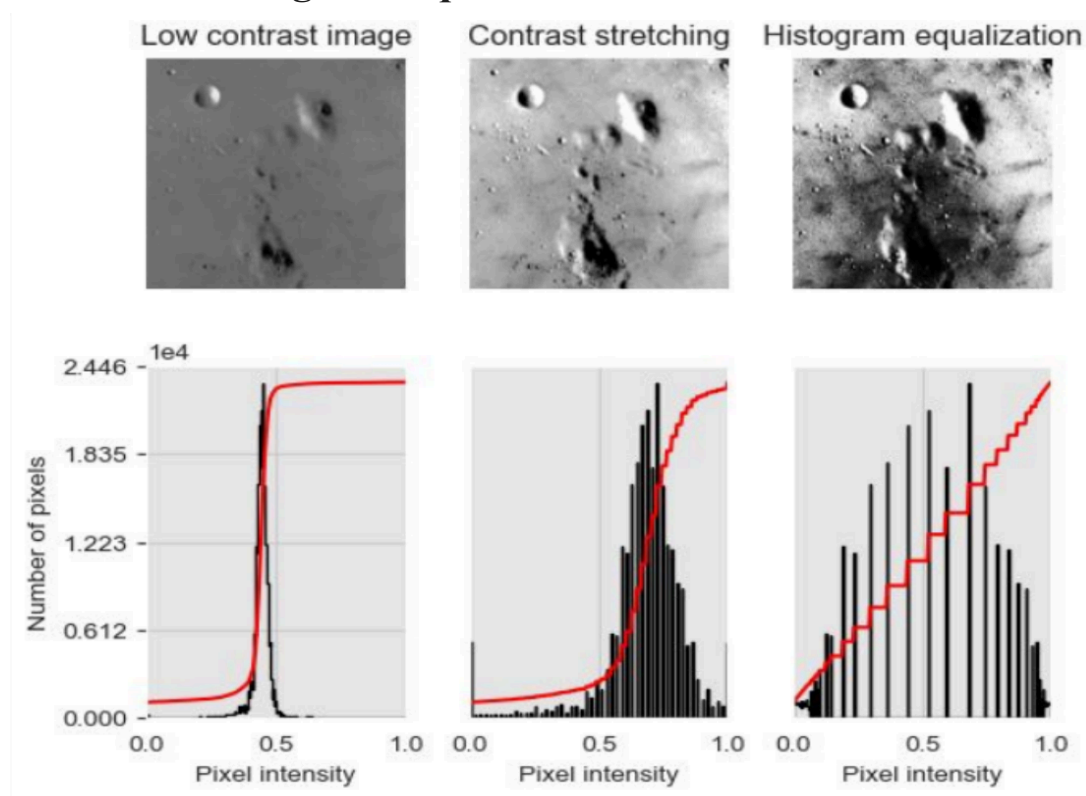
1. Convert RGB to LAB:

- **L (Luminance):** Represents the brightness/intensity.
- **A & B:** Represent the color (chrominance) axes.

2. **Equalize the L Channel Only:** We only apply the histogram equalization to the L channel. This stretches the contrast and improves visibility without touching the color data.
3. **Convert LAB back to RGB:** Once the brightness is adjusted, we convert it back to the standard format for viewing.

Note: By isolating the **Luminance (L)**, we enhance the image's lighting and detail while the **A and B** channels ensure the colors remain identical to the original scene.

Failure of Histogram Equalization



The image explains why **Histogram Equalization (HE)** isn't always the perfect "fix" for image contrast. While it aims to spread out pixel intensities to make an image clearer, it can backfire.

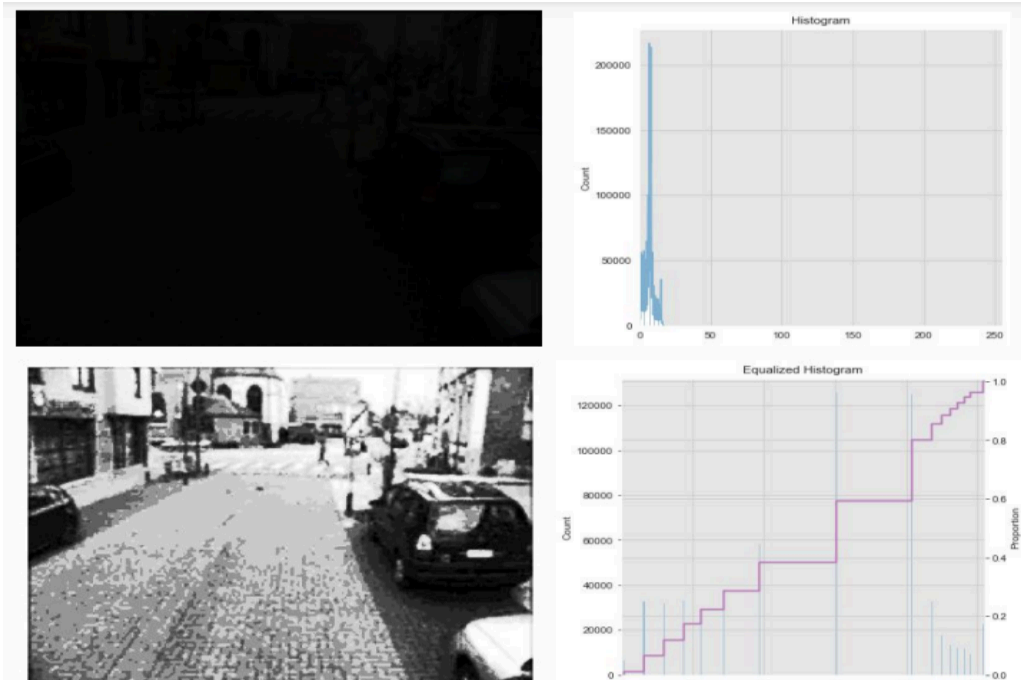
Comparison Breakdown

- **Low Contrast Image:** The original photo is "washed out." All pixels are bunched together in a narrow range of gray (shown by the tall, thin spike in the first graph).
- **Contrast Stretching:** This method simply "pulls" the existing values to the edges. It improves visibility without drastically changing the distribution.
- **Histogram Equalization:** This forces the pixel values to be as uniform as possible across the entire spectrum (0.0 to 1.0). Notice how the third graph is "flattened" and spread out.

Why it "Fails" (The Problems)

1. **Loss of Detail:** By forcing pixels into a uniform distribution, subtle differences in very dark or very bright areas can be merged or flattened, causing features to disappear.
2. **Noise Amplification:** Because HE aggressively stretches the intensity levels, it also stretches the **background noise**. This results in the "grainy" or "gritty" look seen in the third image.
3. **Unnatural Look:** It often creates harsh, artificial-looking lighting that doesn't represent the original scene accurately.

Key Takeaway: Histogram Equalization is great for revealing hidden structures in scientific data, but it often ruins the visual quality of standard photos by making them noisy and overly "harsh."



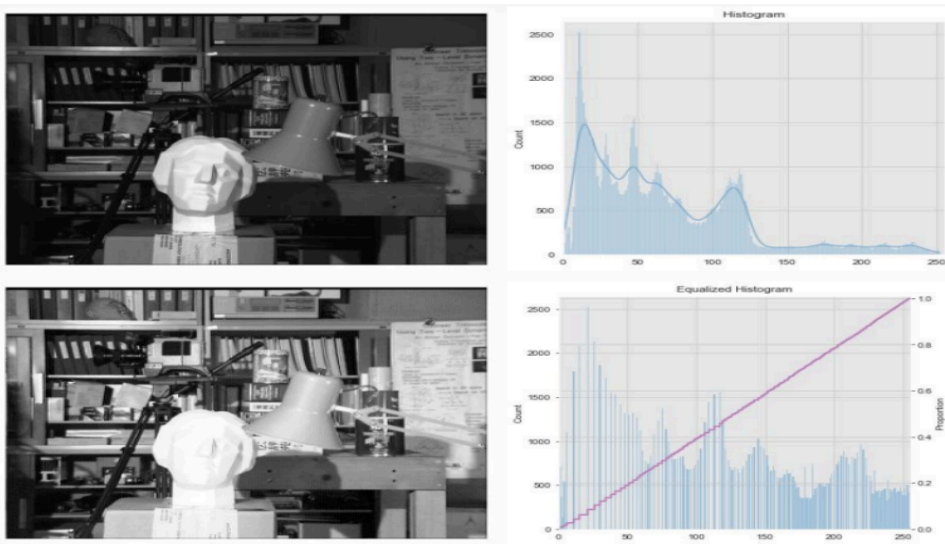
This image illustrates a common limitation of **Histogram Equalization (HE)** in digital image processing. While HE is designed to enhance contrast by spreading out intensity levels, it often fails when an image is dominated by a single, narrow range of values.

Why it Fails here:

- **Intensity Over-concentration:** The original image is very dark, meaning almost all pixels are clustered in the low-intensity (left) side of the histogram.
- **Artifacts and Noise:** When HE tries to stretch this tiny cluster across the full range (0–255), it creates "gaps" in the histogram. This results in the **harsh, washed-out, and grainy** look seen in the bottom-left image.
- **Non-Linear CDF:** In an ideal equalization, the Cumulative Distribution Function (CDF) should be a straight diagonal line. As shown in the "Equalized Histogram" plot, the CDF is a "staircase" instead of a smooth line, indicating that the transformation couldn't achieve a truly uniform distribution.

Summary of Key Points:

- **Low-Intensity Sensitivity:** It performs poorly on images with large, uniform dark or light areas.
- **Loss of Detail:** Instead of bringing out subtle details, it amplifies background noise and creates "patchy" textures.
- **The Alternative:** For images like this, **Adaptive Histogram Equalization (AHE)** or **CLAHE** is usually preferred as they operate on small local tiles rather than the whole image at once.



This slide illustrates a common limitation of **Histogram Equalization (HE)** in digital image processing. While HE is designed to improve contrast by spreading out intensity levels, it doesn't always produce a "better" image.

Key Takeaways:

- **The Mechanism:** HE attempts to create a **flat histogram**, which corresponds to a **linear Cumulative Distribution Function (CDF)**- shown by the purple diagonal line in the bottom graph.
- **The Problem (Over-enhancement):** In the original image, the background was dark and the bust was bright. By forcing a uniform distribution, the algorithm significantly brightened the dark background (making it "clearer"), but at a high cost.

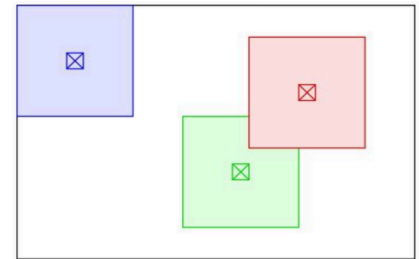
- **Loss of Detail:** Because the algorithm prioritizes global contrast, it "washed out" the subtle gray-level differences on the white bust. The result is a loss of texture and facial features, often referred to as **saturation** or **contouring artifacts**.

Adaptive Histogram Equalization (AHE)

- In some images, **different regions have different brightness** (some very dark, some very bright).
- **Global histogram equalization** may not enhance contrast well in those regions.

Solution (AHE)

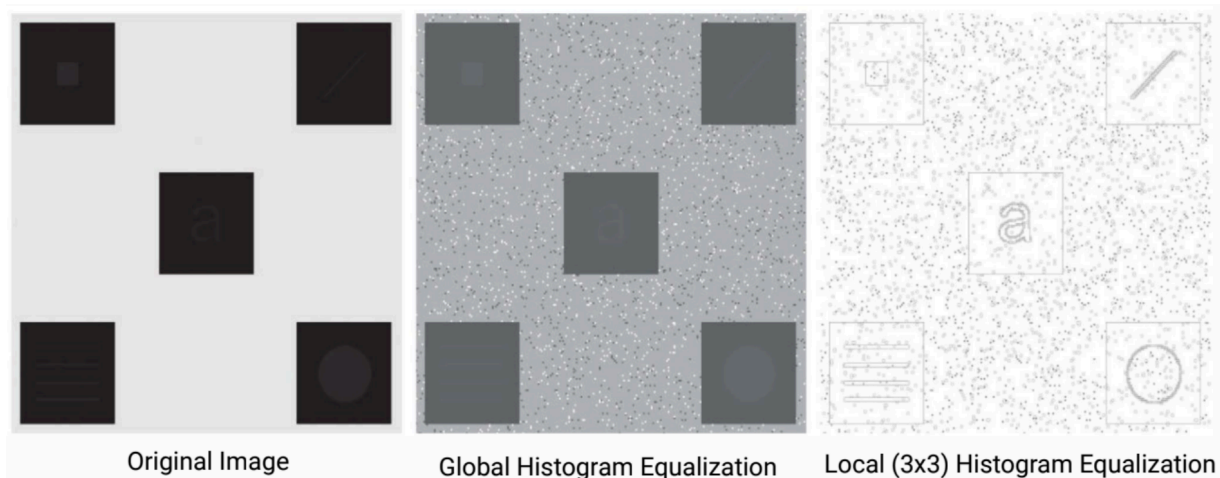
- Divide the image into **small local regions (neighborhoods)**.
- Apply **histogram equalization separately to each region**.



Result

- **Local contrast improves** in both dark and bright areas of the image.

Examples



1. Original Image

The image has extreme contrast. Because the background is very light and the squares are very dark, the global dynamic range is "stretched" to its limits. As a result, the subtle variations inside the dark squares are invisible to the naked eye because they occupy a tiny fraction of the overall histogram.

2. Global Histogram Equalization (GHE)

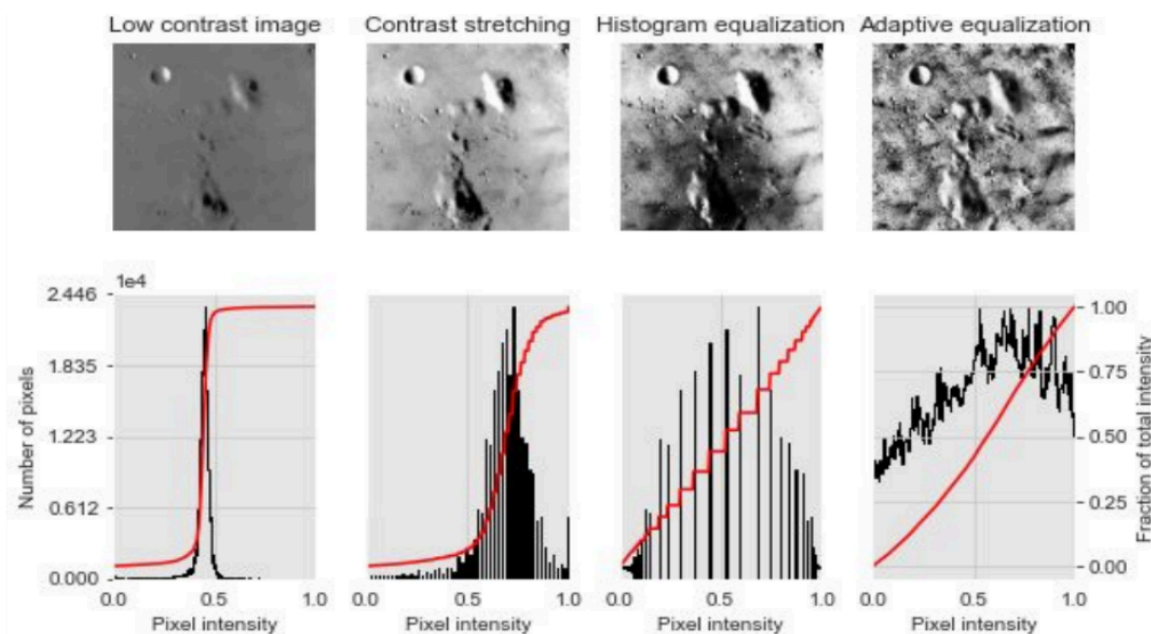
This method calculates one transformation function for the **entire image**.

- **Result:** The image becomes gray and washed out.
- **Why it fails here:** GHE tries to balance the light background and dark squares simultaneously. Because the global distribution is so extreme, it cannot "zoom in" on the small intensity differences within the dark squares. The hidden shapes remain largely invisible.

3. Local (3x3) Histogram Equalization (AHE)

This is the **Adaptive** approach. Instead of one calculation for the whole image, it processes each pixel based on its local neighbors (in this case, a small 3x3 window).

- **Result:** The hidden details- the letter "a," the circle, and the diagonal lines- become clearly visible.
- **How it works:** By looking only at a small neighborhood, the algorithm "ignores" the fact that the rest of the image is bright. Within a dark square, it sees that some pixels are slightly lighter than others and stretches that specific local contrast to fill the full range (black to white).



The image illustrates how different image enhancement techniques redistribute pixel intensities to improve visual contrast. The red line in each graph represents the **Cumulative Distribution Function (CDF)**, which is the key to understanding how these transformations work.

1. Contrast Stretching

- **What it does:** Linearly expands the range of intensity values to fill the available dynamic range (usually 0 to 255).
- **The CDF:** The slope of the CDF becomes steeper in the areas where the original pixels existed, but it remains a relatively simple, piecewise linear mapping.
- **Result:** The image looks clearer, but if the original data was clumped in one spot, it might still look somewhat "flat."

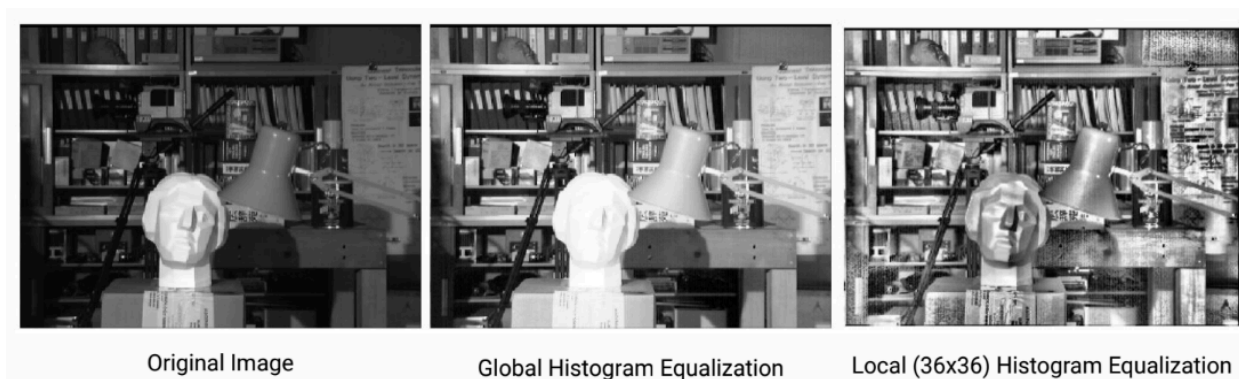
2. Histogram Equalization (HE)

- **What it does:** A global approach that attempts to distribute pixel intensities uniformly across the entire range. It "flattens" the histogram.
- **The CDF:** HE uses the CDF as the transformation function. In an ideally equalized image, the **CDF is a straight diagonal line**, meaning every intensity level has an equal probability of occurring.

- **Result:** Significant contrast boost, but can look "washed out" or introduce noise if certain areas were already bright or dark.

3. Adaptive Histogram Equalization (AHE)

- **What it does:** Instead of looking at the whole image, it divides the image into small tiles (called "contextual regions") and performs equalization on each tile individually.
- **The CDF:** Each tile has its own unique CDF. The algorithm calculates these local CDFs and uses interpolation to eliminate boundaries between tiles.
- **Result:** It excels at bringing out local details that global HE might miss- as seen in the craters and textures of the moon surface in the image- though it can sometimes over-amplify noise in uniform areas (which is why **CLAHE**, or Contrast Limited AHE, is often used instead).



1. Original Image

The image is "low contrast." Most pixels are dark, and the details in the shadows (like the books and the background) are hard to see because the light values are all clumped together.

2. Global Histogram Equalization (GHE)

This method takes the **entire image** and spreads out the most frequent intensity values.

- **Result:** The overall brightness improves, but it applies the same logic to everything.
- **Downside:** It often fails to reveal details in specific dark or bright areas because it's looking at the "average" of the whole picture.

3. Adaptive Histogram Equalization (AHE)

Instead of looking at the whole image, AHE divides the image into several small sections (called **tiles** or "local" regions- in the example, 36x36 pixels).

- **How it works:** It calculates a separate histogram for each tile and redistributes the contrast locally.
- **Result:** Look at the "Local" image on the right. Notice how we can now clearly see the text on the posters and the textures of the books.
- **The "Magic":** Because it operates locally, it can enhance edges and details that are invisible to a global adjustment.

Failure Example



1. Over-Amplification of Noise

In the street scene (the first image), the "Local" result looks very grainy.

- **The Cause:** In areas with very little contrast (like the dark road or the black sky), AHE finds tiny, insignificant variations in pixel intensity and stretches them across the full range of brightness.
- **The Result:** It accidentally turns camera noise or electronic "static" into high-contrast grain that ruins the image.



2. The "Halo" or Checkerboard Effect

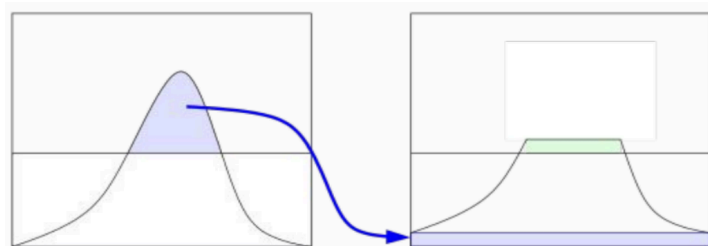
In the moon image (the second image), we can see a strange glow or "aura" around the edges of the moon.

- **The Cause:** Because AHE works in independent tiles (100x100 here), the boundaries between these tiles can become visible.
- **The Result:** The dark background near the moon's edge is treated differently than the dark background further away, leading to artificial gradients and a "patchy" look.

The Solution: CLAHE

To fix this, engineers use **Contrast Limited Adaptive Histogram Equalization (CLAHE)**. It works just like AHE but sets a "limit" on how much contrast can be applied to any single tile. If a tile is too flat (mostly noise), the algorithm clips the histogram to prevent that extreme graininess and haloing we see in the examples.

Contrast Limited Adaptive Histogram Equalization (CLAHE)



Contrast Limited Adaptive Histogram Equalization (CLAHE) is a sophisticated variant of adaptive histogram equalization designed to overcome the limitations of standard global and local methods.



Original Image



Global Histogram Equalization



Local (100x100) Histogram Equalization

Clip limit = 1% of kernel size



Original Image



Global Histogram Equalization



Local (100x100) Histogram Equalization

Clip limit = 1% of kernel size



Original Image

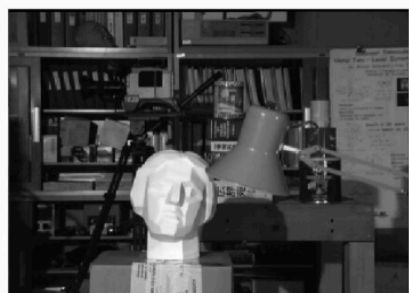


Global Histogram Equalization



Local (100x100) Histogram Equalization

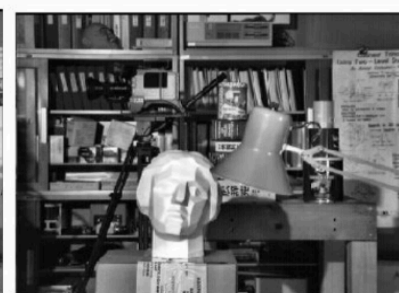
Clip limit = 5% of kernel size



Original Image



Global Histogram Equalization



Local (36x36) Histogram Equalization

Clip limit = 1% of kernel size

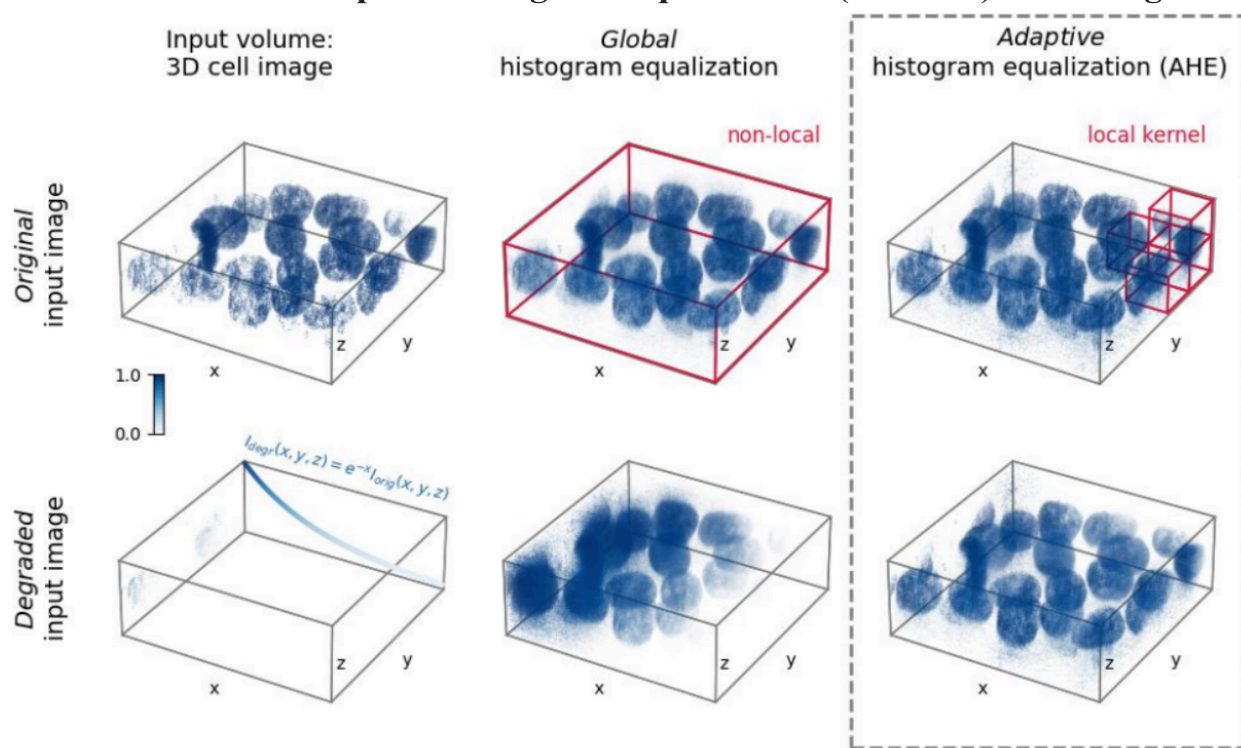
Analysis of Examples

Example	Observation	Why CLAHE helped
Street Scene	The original is almost pitch black.	GHE makes the image grainy and "flat." CLAHE (at 1% clip) recovers the textures of the cobblestones and the car's shape without making the sky look like static noise.
Moon	The original has a narrow dynamic range.	GHE over-brightens the moon, losing crater detail. CLAHE (at 5% clip) brings out the sharp topographical details of the craters while keeping the space background dark.
Desk/Lab	A complex scene with a bright lamp.	GHE "blows out" the white bust near the lamp. CLAHE maintains the detail on the white bust while simultaneously revealing the books on the dark shelves in the background.

Benefits

- **Noise Reduction:** By clipping the histogram, it prevents the amplification of background noise in homogeneous areas.
- **Local Detail:** It excels at revealing "hidden" details in both shadows and highlights within the same image.
- **Natural Appearance:** It produces a much more visually pleasing and realistic result than standard AHE or GHE.

Contrast Limited Adaptive Histogram Equalization (CLAHE) - 3D Image



The image illustrates how different histogram equalization techniques perform on 3D biological data, specifically contrasting **Global** vs. **Adaptive** methods.

1. Global Histogram Equalization (GHE)

- **Mechanism:** It applies a single transformation to the entire image volume based on the global distribution of intensities.
- **Limitation:** As seen in the "Degraded" row, GHE is **non-local**. If the image has **uneven lighting or signal decay** (common in deep tissue imaging), GHE fails to recover details in the darker regions because it treats the whole volume as one unit.

2. Adaptive Histogram Equalization (AHE)

- **Mechanism:** Instead of the whole image, it uses a **local kernel** (the small red cubes in the diagram). It computes histograms for these small sub-regions (tiles) and enhances contrast locally.
- **Strength:** It excels at bringing out detail in heterogeneous images where some parts are much darker than others.

3. Contrast Limited AHE (CLAHE)

- **The "CL" Factor:** Pure AHE often over-amplifies noise in near-constant regions. CLAHE solves this by **clipping the histogram** at a specific value before computing the cumulative distribution function.
- **Result:** It provides the local detail of AHE but with much smoother, more natural-looking results and less noise amplification.

Histogram Matching (Specification)

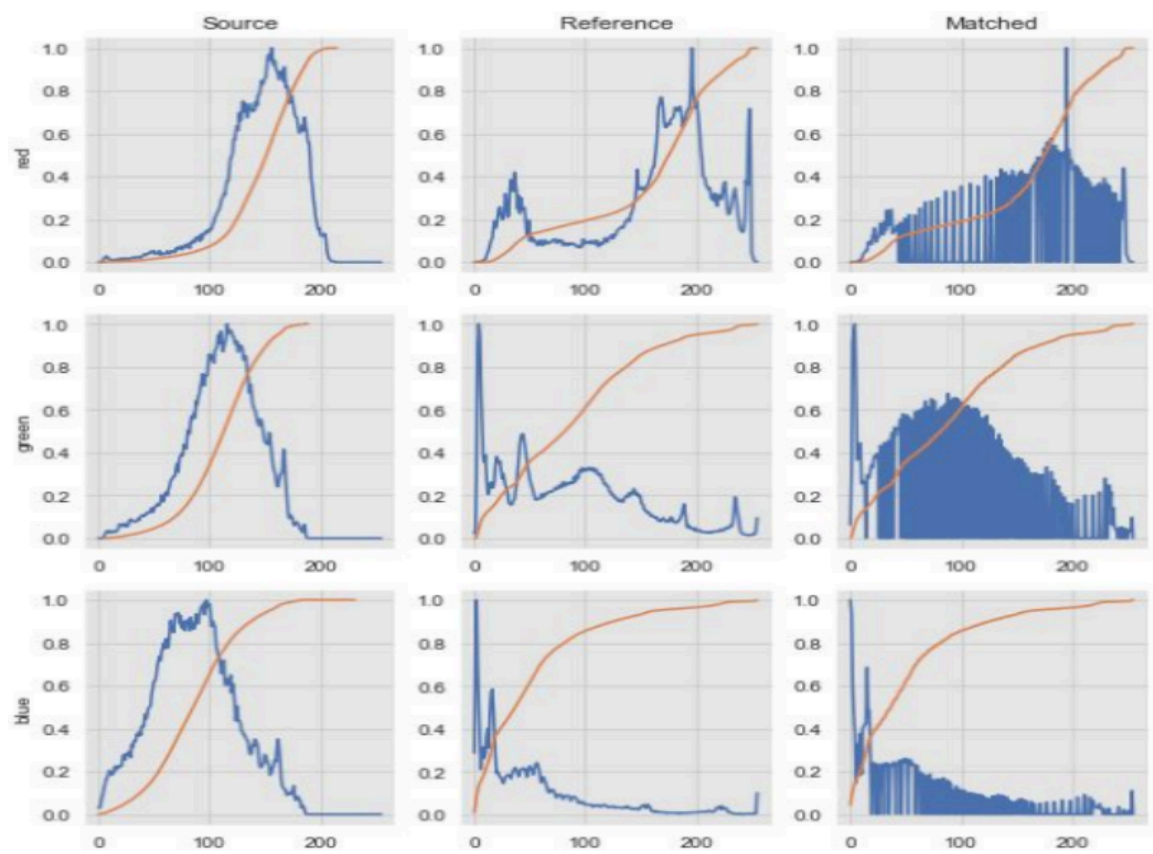


The image illustrates **Histogram Matching** (also known as Histogram Specification), a digital image processing technique used to adjust the pixel intensities of a "Source" image so that its distribution matches that of a "Reference" image.

1. The Core Concept

A histogram represents the frequency of different intensity levels in an image. Matching works by transforming the Source histogram to mimic the Reference histogram's shape.

- **Source:** Has its own unique distribution (e.g., specific lighting/color).
- **Reference:** The "target" style or lighting we want to achieve.
- **Matched:** The result where the cat image now shares the warm, high-contrast tones of the coffee image.



2. Interpreting the Graphs

The right side of the image shows nine plots, organized by color channel (**Red**, **Green**, **Blue**) and process stage:

Column 1: Source Graphs

These show the original distribution of the cat image. Notice the blue "hump" is concentrated in specific areas (mostly mid-tones). The **orange line** is the **Cumulative Distribution Function (CDF)**, which shows the "running total" of pixel frequencies.

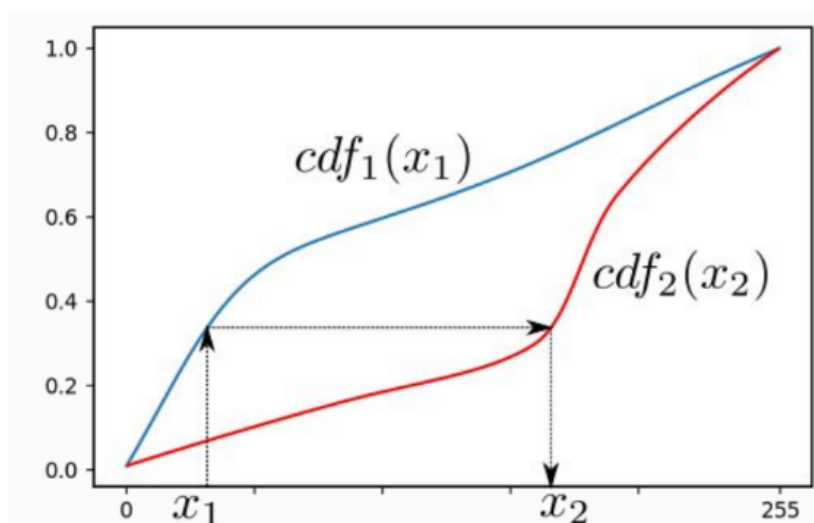
Column 2: Reference Graphs

These represent the coffee cup. The blue histograms are shaped differently- for instance, in the Green and Blue channels, the data is heavily skewed toward the left (darker tones), which gives the coffee image its deep, rich shadows.

Column 3: Matched Graphs

This is the "after" result. Look closely at the blue histograms here: **their overall shape now mimics the Reference column.** * Notice the "comb-like" appearance (vertical lines with gaps). This happens because the original discrete pixel values are being stretched or squeezed to fit the new shape.

- The **orange CDF lines** in the Matched column now almost perfectly match the slope and curve of the Reference CDF lines.



This image illustrates the mathematical process for performing **Histogram Matching** by using the **Cumulative Distribution Function (CDF)** of two images.

Breakdown of the Graph

The graph shows two CDF curves plotted against pixel intensity (ranging from **0** to **255**).

- **Blue Curve (cdf_1):** This represents the cumulative distribution of the **Source** image.
- **Red Curve (cdf_2):** This represents the cumulative distribution of the **Reference** (target) image.
- **The Mapping Path:** The dotted horizontal line shows that for a specific input intensity x_1 , we find its corresponding cumulative probability on the blue curve, then move horizontally to find the intensity x_2 on the red curve that shares that same probability.

Explaining the Formula

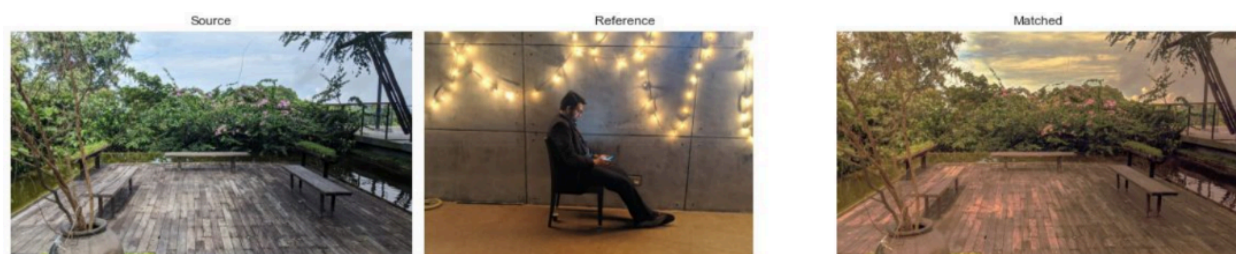
The equation is the formal way to describe this "look-up" process:

$$x_2 = T(x_1) = cdf_2^{-1}(cdf_1(x_1))$$

- **cdf_1(x_1)**: take a pixel value x_1 from the source image and find its percentile (its "rank") in the source distribution.
- **cdf_2⁻¹**: take that same percentile and apply the **inverse** CDF of the reference image. This asks: *"What intensity value in the reference image corresponds to this specific percentile?"*
- **x₂**: The result is the new pixel value for the matched image.

Example

Imagine x_1 is a medium-gray pixel in a very dark photo. Its $cdf_1(x_1)$ might be **0.9** (meaning it's brighter than 90% of the other pixels). If the reference image is very bright, its cdf_2 curve will be shifted right. To find the new value x_2 , we look for the pixel value that is brighter than 90% of the pixels in that bright reference image.



How it Works

1. **Source:** The original image (the outdoor deck) has its own lighting and color profile.
2. **Reference:** This image (the person under warm lights) serves as the "template" for the desired look.
3. **Matched:** The algorithm calculates a mathematical transformation so that the **Source** image's histogram matches the **Reference** image's histogram.

Prepared By Azmari Sultana ✨